In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II
(PAPER-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	B
2.	D	D	A
3.	C	C	C
4.	C	C	C
5.	C	B	B
6.	A	D	B
7.	B	C	D
8.	C	D	A
9.	ABC	ABD	AB
10.	A	BC	AD
11.	ABCD	ABD	ABC
12.	ACD	AC	AC
13.	A	A	C
14.	C	A	D
15.	A	B	C
16.	D	C	A
17.	A	C	C
18.	D	C	B
1.	2	7	3
2.	6	4	2
3.	4	4	3
4.	4	5	6
5.	2	4	4

Physics

PART – I

SECTION – A

1. In the first case, m will remain at rest.

$$a_M = \frac{F}{M}$$

In the second case, both will accelerate:

$$a_m = a_M = \frac{F}{M+m}$$

In the second case, force on $m = ma_m = \frac{mF}{M+m}$

2. $k_1x_1 = k_2x_2$

as $k_2 = 3k$

$$\frac{x_1}{x_2} = 3 \text{ and } \frac{\frac{1}{2}k_1x_1^2}{\frac{1}{2}k_2x_2^2} = \frac{x_1}{x_2} = 3$$

3. $$N = \frac{mv^2}{r}; f = \mu N = mg$$

$$\text{or } \frac{\mu mv^2}{r} = mg$$

$$v = \left(\frac{gr}{\mu}\right)^{1/2}$$

4. Thermometer, ice cream box and freezer are in thermal equilibrium.

6. $$R = \frac{20^2 \times \sin 120°}{g} = 20\sqrt{3} = \frac{\Delta R}{R} = \frac{24U}{U} \Rightarrow \Delta R = \frac{2\times 5}{100} \times 200\sqrt{3} = 2\sqrt{3}$$

$$20\sqrt{3} - 2\sqrt{3} < R < 20\sqrt{3} + 2\sqrt{3} \Rightarrow 31.1\text{m} < R < 38.1\text{m}$$

7. $$\left(\frac{\in_0 klx}{d} + \frac{\in_0 (2A - lx)}{d}\right)V = Q$$

$$i = \left(\frac{\in_0 klx}{d} + \frac{-\in_0 l}{d}\right)\frac{dx}{dt} = \text{constant}$$

As the dielectric levels $c \downarrow \Rightarrow Q \downarrow \Rightarrow i$ is – ve

8. $\Delta\phi = 2n\pi$

$$\Rightarrow \frac{\pi}{2}+\frac{2\pi}{\lambda}d\sin\theta = 2n\pi$$

$$\frac{2\pi}{\lambda}d\sin\theta = \left(2n-\frac{1}{2}\right)\pi$$

$$\sin\theta = \left(2n-\frac{1}{2}\right)\frac{\lambda}{2d} = \frac{1}{2}\times\frac{\lambda}{2\times 3\lambda} = \frac{1}{12}$$

$$\Rightarrow \frac{y}{\sqrt{(100\lambda)^2}} = \frac{1}{2}$$

$$144y^2 = (100\lambda)^2$$

$$y \approx \frac{100\lambda}{12} = \frac{25\lambda}{3}$$

9. $V_A = V_B \Rightarrow \text{work} = 0$

10. Radius $R_1 = \dfrac{nv}{qB_1}$

In a given field radius should be same for every entry. So B & C are not possible.

15. $$f = \frac{C-V_R}{C+V_R}f_0$$

$$\frac{\Delta f}{f_0} = \frac{2V_R}{C+V_R} \approx \frac{2V_R}{C}$$

$$\Rightarrow C = \frac{2\times 0.1}{600}\times 5\times 10^6 \approx 1700\,\text{m/s}$$

16. $$f' = \frac{C-V_R}{C+V_R}f_0$$

$$\frac{\Delta f}{f_0} = \frac{2V_R}{C+V_R}$$

Velocity is 4 times $\Rightarrow \Delta f = 4\Delta f_0 = 2400\,\text{Hz}$

17. Reading of C = V $\{i \text{ in that branch} = 0\}$

Reading of $A = \dfrac{V}{R}$

$\Rightarrow R$

18. $$V_B = iR_{\text{ammeter}} = 0$$

$$\Rightarrow \quad V_{cap} = 0$$

$$\Rightarrow T = 2\pi\sqrt{LC} = 2\pi\sqrt{\frac{1}{10}} = 2\,\text{sec}$$

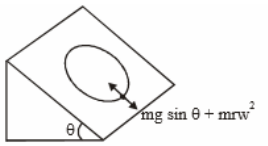
SECTION – C

1. $P = 700\times10^{3}\times1.6\times10^{-19}\times\dfrac{dn}{dt} = 10\times10^{-3}$

$\dfrac{dN}{dt} = \dfrac{10^{-1}}{10^{-14}}\times\dfrac{1}{7\times16} = \dfrac{10^{12}}{11.2} = \lambda N_0$

$\lambda = \dfrac{\ln 2}{14\times86400} \Rightarrow N_0 = \dfrac{14\times86400\times10^{12}}{11.2\ln 2} = 154\times10^{15}$

2. $f \geq mg\sin\theta + mrw^2$



$\mu mg\cos\theta \geq mg\sin\theta + mrw^2$

or $\mu \geq \tan\theta + \dfrac{w^2 r}{g\cos\theta}$

3. $d\phi = \dfrac{\mu_0 i}{2\pi x}\times l\,dx$

$\phi = \dfrac{\mu_0 i}{2\pi} l\ln\left(\dfrac{d_2}{d_1}\right)$

$\varepsilon = \dfrac{d\phi}{dt} = \dfrac{\mu_0}{2\pi} l\ln\left(\dfrac{d_2}{d_1}\right)\times i_0\omega\cos wt$

$\varepsilon_{rms} = \dfrac{\mu_0}{2\pi} l\ln\left(\dfrac{d_2}{d_1}\right)\dfrac{i_0\omega}{\sqrt{2}}$

$= \dfrac{4\pi\times10^{-7}}{2\pi}\times l\times\ln 2\times\dfrac{2\sqrt{2}}{\ln 2}\times\dfrac{50}{\sqrt{2}}$

4. The block slips when the static friction force reaches its maximum value $f_{s\,max} = \mu_s n = \mu_s mg$ .This occurs when the train has moved

$\Delta x = \dfrac{f_{smax}}{k} = \dfrac{\mu_s mg}{k} = \dfrac{(0.60)(2.0kg)(10m/s^2)}{50N/m} = 24\,cm$

The time at which the block slips is

$t = \dfrac{\Delta x}{v} = \dfrac{24cm}{6\,cm/s} = 4s$

5. $\tau_1 = RC$

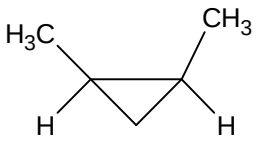
$\tau_2 = \dfrac{L}{R}$

$\Rightarrow LC = t_1 t_2 = 0.1sec$

Chemistry

PART – II

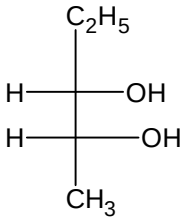
SECTION – A

1. 

2. $BaTiO_3$

3. $(NH_4)_2SO_4$, $(NH_4)_2CO_3$, NH_4Cl all give NH_3 on heating

4. ΔG is the thermodynamical concept

5. 

+ Mirror image

6. β-Keto acid on heating shows de-carboxylation on heating, through enol form

7. Ca^{2+} has 18 e^- and Cl^- has 18 e^-.

8. Phenol gives 2,4,6-tribromo phenol with aq. Br_2.

9. BF_4^- is tetrahedral and rest are is square planar

10. $Z = \frac{V_{m,real}}{V_{m,ideal}}$

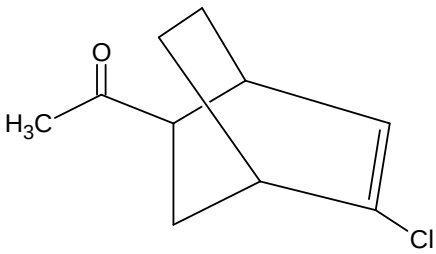
at high temperature and low pressure the gas follows PV=nRT

11. $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$

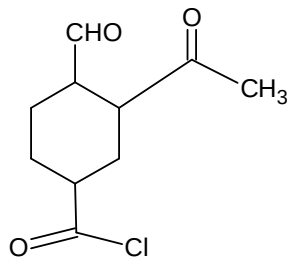
$2Na + H_2SO_4 \longrightarrow Na_2SO_4 + H_2$

$2Al + 6H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3SO_2 + 6H_2O$

15. CCl_2 will form and show addition reaction then ring expansion takes place.

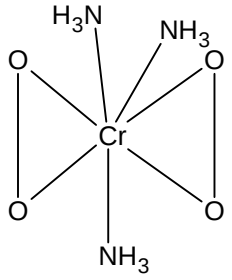
16. 

(B)



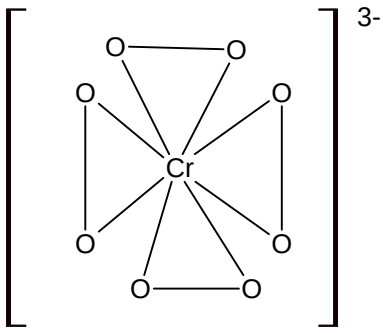
(C)

17.



Pentagonal bipyramid

18.



Pentagonal bipyramid

SECTION – C

1.

$$\frac{dq}{dT}=\frac{\Delta U\text{-}w}{ndT}=C_{m,V}+\frac{PdV}{dT}=\frac{3}{2}R+\frac{P}{n}\frac{dV}{dT}\text{-----}(1)$$

$$PV=nRT$$

$$PdV+VdP=nRdT$$

$$P+\frac{VdP}{dV}=nR\frac{dT}{dV}$$

$$P+V(-1)=nR\frac{dT}{dV}$$

$$4+2(-1)=nR\frac{dT}{dV}$$

$$\frac{dT}{dV}=\frac{2}{nR}$$

$$\frac{dV}{dT}=\frac{nR}{2}$$

Put in (1)

$$\frac{dq}{dT}=\frac{3}{2}R+\frac{P}{n}\frac{nR}{2}$$

$$\frac{dq}{dT}=\frac{3}{2}R+\frac{P}{}\frac{R}{2}=\frac{3}{2}R+\frac{4}{}\frac{R}{2}=3.5R$$

2. d,e,f,g are correct and Colloid is +ve.

3. Volume of unit cell= $\frac{\sqrt{3}}{4}\times(4.53)^2\times7.41\times10^{-24}cm^3$

so mass=60.57×10^{-24}g

moles= 3.365×10^{-24}

molecules of H_2O=2.025

number of H atom=4

4. $K_{In}=\frac{\left[H^+\right]\left[I_n^-\right]}{\left[HI_n\right]}$

$=10^{-9.18}\times\frac{0.6}{0.4}$

$=\frac{3}{2}\times10^{-9.18}$

$K_{In}=10^{-9}\frac{\left[In^-\right]}{\left[HI_n\right]}$

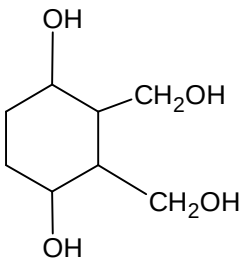
$K_{In}=\frac{3}{2}\times10^{-9.18}=10^{-9}\times\frac{\alpha}{1-\alpha}$

$\frac{3}{2}\times0.66=\frac{\alpha}{1-\alpha}$

$\frac{\alpha}{1-\alpha}=1$

$2\alpha=1$

$\alpha=0.5$

5. 

(A)

Mathematics PART – III

SECTION – A

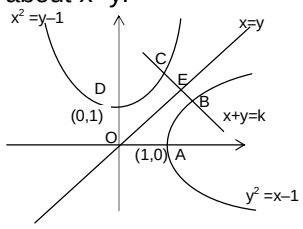
2. As s–b, s–c, s–a are in AP with common difference =1
$\Rightarrow$ a,c, b are in AP with common difference equal to 1.

$\Rightarrow$ c = a+ 1, b = a+2 $\quad\Rightarrow s = \dfrac{a+a+1+a+2}{2} = \dfrac{3(a+1)}{2}$

$$\Rightarrow \Delta = \sqrt{\frac{3}{2}(a+1).\frac{a+3}{2}.\frac{a+1}{2}.\frac{a-1}{2}}$$

Also $r = \Delta/s \Rightarrow a^2 + 2a - 195 = 0 \Rightarrow (a+15)(a-13)=0$
A = 13, c=14, b=15 $\quad\Rightarrow \Delta = 84$ sq. units

3. Line of shortest distance will be perpendicular to line x=y. Also both parabolas are symmetric about x=y.



A=area OABCDO
A′=area OECDO
$\Rightarrow$ A′ =A/2

4. Let α be the real root then $(a\alpha^2 + 2b\alpha + c) + i(d\alpha^2 + 2e\alpha + f) = 0$

$$\Rightarrow a\alpha^2 + 2b\alpha + c = 0 \Rightarrow \alpha = \frac{-\sqrt{c}}{\sqrt{a}}$$

$$\Rightarrow d\alpha^2 + 2e\alpha + f = 0 \qquad \Rightarrow d.\frac{c}{a} - 2e\frac{\sqrt{c}}{\sqrt{a}} + f = 0 \qquad \Rightarrow \frac{d}{a} - 2.\frac{e}{b} + \frac{f}{c} = 0$$

5. $I = \int \tan^{-p/3} x.\sec^4 x\,dx$
$= \int \tan^{-p/3} x(1+\tan^2 x).\sec^2 x\,dx$

6. If equation of line AB is $\vec{r} = \vec{0} + \lambda(\sqrt{2}a\hat{i} - \vec{0}) = \lambda\hat{i}$
then equation of line DC is $\vec{r} = \sqrt{2}a\hat{i} + \mu\left(\frac{a}{\sqrt{2}}\hat{i} + \frac{a}{\sqrt{2}}\hat{j} + a\hat{k} - \sqrt{2}a\hat{i}\right)$

$$\vec{r} = \sqrt{2}a\hat{i} + \mu\left(\frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \hat{k}\right)$$

$$\Rightarrow \quad S.D = \left|\frac{(\vec{a}_2 - \vec{a}_1).(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}\right| = \frac{2a}{\sqrt{3}}$$

7. $$Q = \int_0^\infty \frac{x\,dx}{1+x^4} = \frac{1}{2}\left|\tan^{-1} x^2\right|_0^\infty = \frac{1}{2}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{4}$$

$$P=\int_0^{\infty}\frac{x^2dx}{1+x^4},\ \text{Put } x=\frac{1}{t}\Rightarrow \qquad P=\int_{\infty}^{0}\frac{\frac{1}{t^2}(\frac{-1}{t^2}dt)}{1+\frac{1}{t^4}}=\int_0^{\infty}\frac{dt}{1+t^4}=R$$

$$\text{Also } P+R=\int_{\infty}^{0}\frac{x^2+1}{1+x^4}dx=\int_0^{\infty}\frac{(1+\frac{1}{x^2})dx}{x^2+\frac{1}{x^2}}=\frac{\pi}{\sqrt{2}}$$

8. $\int_{\pi/2}^{x}\sqrt{3-2\sin^2 t}+\int_0^{y}\cos t\,dt=0$, then $\left(\frac{dy}{dx}\right)$ at $x=\pi$ and $y=\pi$

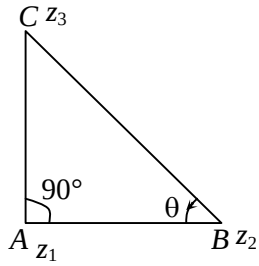
$$\left(\sqrt{3-2\sin^2 t}\right)\cdot 1+\cos y\frac{dy}{dx}=0$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{\pi,\pi}=\left.\frac{-\sqrt{3-2\sin^2 x}}{\cos y}\right|_{\pi,\pi}=\sqrt{3}.$$

9. $$\frac{W_2}{|W_2|}=\frac{W_1}{|W_1|}e^{-i\theta}$$

$$\Rightarrow \qquad \frac{W_2}{W_1}=\frac{|W_2|}{|W_1|}e^{-i\theta}=1-i\tan\theta$$

$$W_1\overline{W}_2=\frac{W_1}{W_2}|W_2|^2=|W_1||W_2|e^{-i\theta}=\left(|W_1|^2-i|W_1||W_2|\sin\theta\right)$$



10. $(1+x+x^2)^n=a_0+a_1x+a_2x^2+\ldots+a_{2n}x^{2n}$ ...(1)

$$\Rightarrow a_0+a_2+a_4\ldots=\frac{3^n+1}{2}$$

$$a_1+a_3+a_5\ldots=\frac{3^n-1}{2}$$

Put $x=i$ in equation (1) where $i=\sqrt{-1}$

If $n=4m+1$

$\Rightarrow$ $a_1-a_3+a_5\ldots=1$

$a_0-a_2+a_4\ldots=0$

So, $a_1+a_5+a_9\ldots=\frac{3^n+1}{4}=S_2$

$a_3+a_7+a_{11}\ldots=\frac{3^n-3}{4}=S_4$

Similarly if $n=4m+3$

Then, $S_2=\frac{3^n-3}{4}$ and $S_4=\frac{3^n+1}{4}$

11. If a,b,c are odd integer then D can not be perfect square,

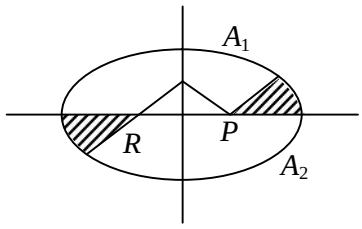
$\Rightarrow$ $|f(\lambda)|>0$ where $\lambda\in Q$

$\Rightarrow \quad |\, a\,p^2 + b\,pq + c\,q^2\,| \geq 1 \;\Rightarrow \quad \left| f\left(\frac{p}{q}\right)\right| \geq \frac{1}{q^2}$

If $\quad a.c = 1 \quad \Rightarrow \quad a = c = 1$ and $b \geq 3$ as

$\quad c.(a - b + c) < 0 \quad \Rightarrow \quad \alpha \in (-1, 0)$

12. Area of ellipse $= \pi \times \frac{8}{3\sqrt{A}} \times \frac{5}{2\sqrt{A}} = \frac{20}{3}$



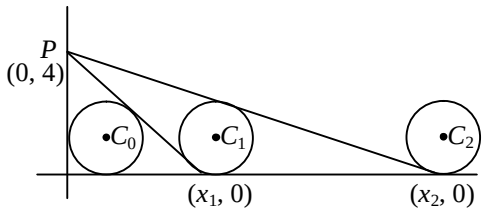
Both shaded areas are equal, then

$$A_1 = \frac{1}{2}Ae - \Delta PQR = \frac{10}{3} - 1$$

$$A_1 = \frac{1}{2}Ae + \Delta PQR = \frac{10}{3} + 1$$

(Q. No. 13 – 14)

For C_n circle let centre be $(\alpha_n, 1)$ and it touches line L_n and L_{n+1}.



i.e. $\quad \frac{x}{x_n} + \frac{y}{4} = 1,\ \frac{x}{x_{n+1}} + \frac{y}{4} = 1$

$\Rightarrow \quad \frac{4\alpha_n + x_n - 4x_n}{\sqrt{16 + x_n^{\,2}}} = 1 \qquad \ldots(i)$

$\Rightarrow \quad \frac{4\alpha_n + x_{n+1} - 4x_{n+1}}{\sqrt{16 + x_{n+1}^2}} = -1 \qquad \ldots(ii)$

$\Rightarrow \quad 3(x_{n+1} - x_n) = \sqrt{16 + x_n^2} + \sqrt{16 + x_{n+1}^2}$

Let $\quad x_n = 4\cot\theta_n \; 0 < \theta_n < \frac{\pi}{2}$

$\Rightarrow \quad \tan\left(\frac{\theta_{n+1}}{2}\right) = \frac{1}{2}\tan\left(\frac{\theta_n}{2}\right)$

As $\quad x_1 = 3 \;\Rightarrow\; \tan\frac{\theta_1}{2} = \frac{1}{2} \;\Rightarrow\; \tan\left(\frac{\theta_n}{2}\right) = \frac{1}{2^n}$

$\Rightarrow \quad x_n = \left(\frac{2^{2n} - 1}{2^{n-1}}\right) \qquad x_2 = \frac{15}{2}$

Putting is (i), we get $\alpha_2 = \frac{31}{4}$.

$$\lim_{n\to\infty} \frac{x_n}{2^n} = \lim_{n\to\infty} \frac{2^{2n} - 1}{2^{n-1}2^n} = 2$$

15. According to paragraph $\frac{f(b) - f(a)}{b - a} > f'(c), \frac{g(b) - g(a)}{b - a} = g'(c)$ and $\frac{h(b) - h(a)}{b - a} < h'(c)$

16. If $g(x) = Ax^2 + Bx + C$

$$\Rightarrow \frac{g(b)-g(a)}{b-a}=\frac{A(b^2-a^2)+B(b-a)}{b-a} \Rightarrow 2A\frac{(b+a)}{2}+B=g'\left(\frac{b+a}{2}\right)$$

17. Put $x = y = 0$ in $f(x + y^3) = f(x) + (f(y))^3$

$f(0) = 0$

$$\Rightarrow k = \lim_{x\to 0}\frac{f(h)-f(0)}{h}=\lim_{x\to 0}\frac{f(h)}{h}=\lim_{x\to 0}\frac{f\left(0+\left(h^{1/3}\right)^3\right)-f(0)}{\left(h^{1/3}\right)^3}$$

$$=\lim_{x\to 0}\left(\frac{f\left(h^{1/3}\right)}{\left(h^{1/3}\right)}\right)^3=k^3 \Rightarrow k = 0, 1, -1$$

since $|k| > 0 \Rightarrow k = 1, -1$

$$\Rightarrow f'(x)=\lim_{x\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{x\to 0}\left(\frac{f\left(h^{1/3}\right)}{\left(h^{1/3}\right)}\right)^3=k^3$$

$\Rightarrow f'(x) = -1$ or $1 \ \forall\ x \in R$

$\Rightarrow f(x) = -x + c$ or $x + c \ \forall\ x \in R$

$\because f(0) = 0$

$\Rightarrow f(x) = -x \ \forall\ x \in R$

or $f(x) = x \ \forall\ x \in R$

$\frac{\tan x}{x} > 1 \ \forall\ x \in (-h, h)$

and $\frac{\tan x}{-x} < -1 \ \forall\ x \in (-h, h)$

hence $\lim_{x\to 0}\left[\frac{\tan x}{f(x)}\right] = 1$ or -2.

18. If $k = 0 \Rightarrow f'(0) = 0$

$$\Rightarrow f(x)=\lim_{x\to 0}\left(\frac{f\left(h^{1/3}\right)}{\left(h^{1/3}\right)}\right)^3=0 \Rightarrow f(x) = c$$

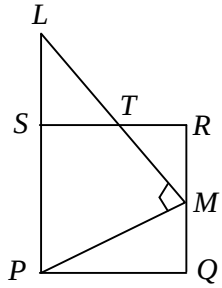
since $f(0) = 0 \Rightarrow f(x) = 0 \ \forall\ x \in R$.

SECTION – C

1. Let $Q(ai),\ S(bj),\ M\left(ai+\frac{b}{2}j\right), L\left(\frac{4}{3}bj\right),\ R(ai+aj)$

be required points:

T is intersection of LM and SR



i.e. $ai+\frac{b}{2}j+\lambda\left(ai+\frac{b}{2}j-\frac{4}{3}bj\right)=bj+\mu(ai)$

$\lambda=\frac{-3}{5}$ and $\mu=\frac{2}{5}$ one comparing the coefficient, we get $T\equiv\left(bj+\frac{2a}{5}i\right)$

Area of $\Delta MRT=\frac{1}{2}\times\frac{3.a}{5}\times\frac{b}{2}=\frac{3}{20}ab$.

2. Let first equation have roots α, β, γ_1 and second equation α, β, γ_2

then $\alpha+\beta+\gamma_1=-5, \qquad \alpha\beta+\gamma_1(\alpha+\beta)=p$

$\alpha+\beta+\gamma_2=-7, \qquad \alpha\beta+\gamma_2(\alpha+\beta)=p$

$\Rightarrow (\gamma_1-\gamma_2)(\alpha+\beta)=0$ as $(\gamma_1\neq\gamma_2)\Rightarrow\alpha+\beta=0$

$\gamma_1=-5,\ \gamma_2=-7$.

3. $(23)^{23}=(23)(530-1)^{11}=53\lambda-23$

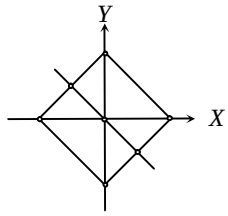
Hence remainder is 30.

4. $\sin x+\sin y=\sin(x+y)$

$\Rightarrow \sin\left(\frac{x+y}{2}\right)\sin\frac{x}{2}\sin\frac{y}{2}=0$

$\Rightarrow x+y=2h_1\pi,\ x=2h_2\pi$ or $y=2h_3\pi,\ h_1, h_2, h_3\in Z$

Only six solutions are possible



5. $\left\{\frac{4n^4}{4n^2-1}\right\}=\left\{\frac{1}{4}\left(4n^2+1+\frac{1}{4n^2-1}\right)\right\}$

$=\sum_{n=1}^{16}\left(\frac{1}{4}+\frac{1}{8}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right)$

$=4+\frac{1}{8}\left(1-\frac{1}{33}\right)$

$=4+\frac{4}{33}$

Hence, $\left[33\left\{4+\frac{4}{33}\right\}\right]=4$.